

What is CPS?

Electrical Power Grid

Why is this a CPS?

Electrical power grids are a CPS because it has both a physical system and a cyber system. Both these systems work together to transfer the electric voltage to its customers.

The physical system consists of the power stations, substation, power transmission, power lines, and power distribution. This is what we can see.

The cyber system consists of how the power station, substation and power distribution operate. Their task is to monitor the electric voltage traveling through the grid and make sure it isn't lost and distribute the electricity such that the customer gets what it was promised.

The power station is the station that generate electricity. It sends it to a step-up transformer which takes the voltage up so it can travel greater distance. Before being distributed to customers, it goes through a step-down transformer which takes it back down to normal. After that it gets sent to customers through distribution.

Self-driving cars

Why is this a CPS?

A self-driving car consists of the physical system which is the car and the engine, and the cyber-system which consist of robotic and advanced sensors which makes the car not needing human input.

Why is a computer scientist needed for the modelling, analysis, and design?

A computer scientist might be needed because to model such systems, you need a broad understanding of how I/O control unit and sensors work together. With this knowledge, a computer scientist can create a model that defines the structure of the system and how it will work.

With a model created, a computer scientist can start the design process and determine what types of sensors one might need and will get to a deeper level of understanding how the physical system will work with the cyber-system.

Lastly, we need an analysis of the system that is created.

These analyses consist of:

- Invariants and Temporal Logic – precise descriptions of dynamic properties of systems
- Equivalence and Refinement – relationship between models
- Reachability analysis and Model checking – analyze large number of possible behaviors that a model may exhibit
- Quantitative analysis – analyze quantitative properties of embedded software – time analysis (energy and memory too)
- security and privacy – basics of security and privacy for embedded systems design.